USER INTERFACE AND DESIGN

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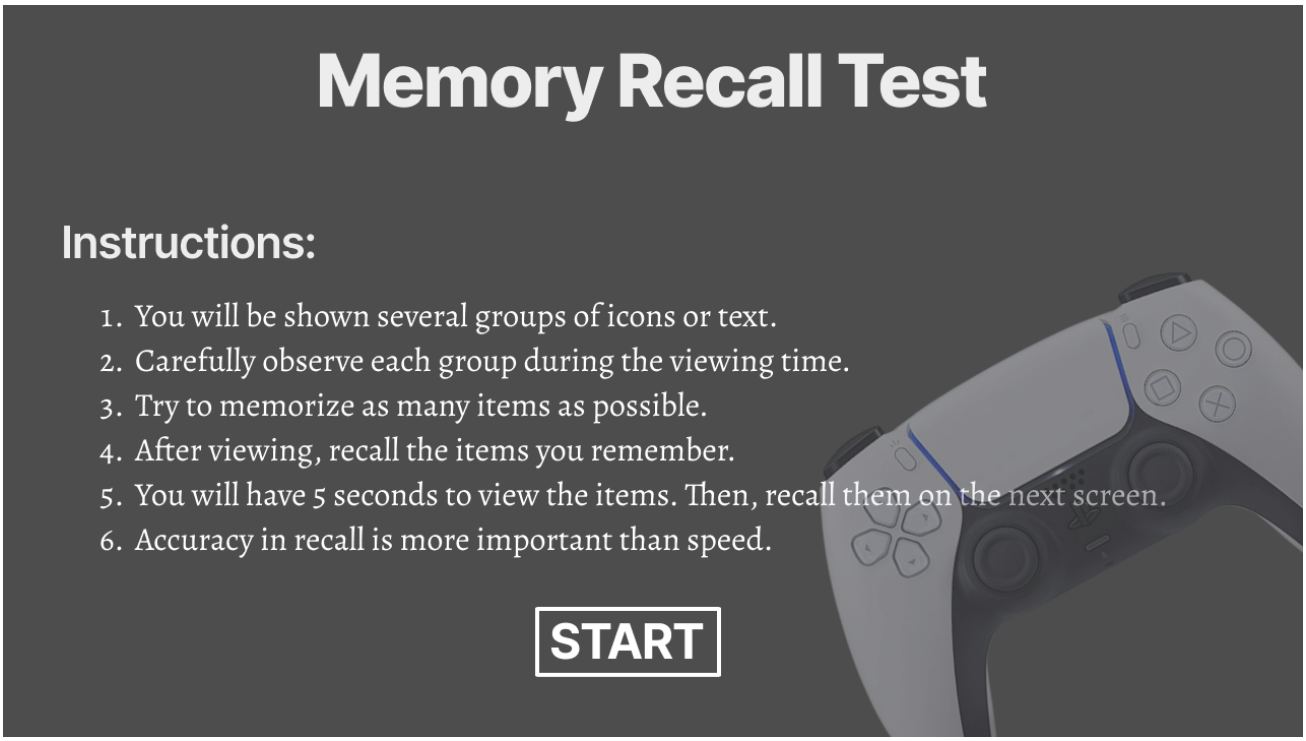
Exp-2: Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1 (INSTRUCTION PAGE):

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:



1. Clear and Sequential Numbering:

- The instructions are broken down into six steps, making it easier to follow.
- Each step presents one key action in a structured manner, reducing cognitive overload.

2. Logical Grouping of Information:

• Observation Phase (Steps 1-2) → Users learn about what they will see.
• Memorization Strategy (Step 3) → Encourages users to focus on remembering items.
• Recall Phase (Steps 4-5) → Explains how users will recall information.
• Key Performance Metric (Step 6) → Highlights that accuracy matters more than speed.

3. Visual Hierarchy and Design Elements:

• Bold, large title ("MEMORY RECALL TASK") → Grabs attention and clearly states the task.
• Bullet points and spacing → Reduce clutter, enhancing readability.
• Highlighted "START" button → Signals the next step, keeping navigation intuitive.
• Bee-themed visuals → Add a playful touch, engaging users without distraction.

4. Time Constraint Reinforcement:

• Step 5 explicitly states, “You will have 5 seconds to view the items.”
• This reinforces expectations while subtly urging users to focus.

5. Simplicity and Clarity:

• Instructions use short, direct sentences, avoiding unnecessary complexity.
• The active voice makes it more engaging and action-oriented.

<u>FRAME 2 (CHUNKING PHASE</u>):

This screen represents the Chunking Phase of a Memory Recall Task, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

1. Purpose of the Screen:

- This is the visual memory encoding phase, where users view and group items mentally before recalling them.
- The term "Chunking Phase" suggests that the game applies the chunking principle, which helps users remember items by categorizing them into meaningful groups.

2. Key Elements and UI Components:

3. How the Chunking Phase Works:

- Users scan the grid and look for patterns or related items to create mental chunks.
- The countdown timer limits observation time, forcing quick memory strategies.

• Once time is up, users transition to the recall phase, where they must identify previously seen items.

4. Cognitive and UX Benefits of Chunking:

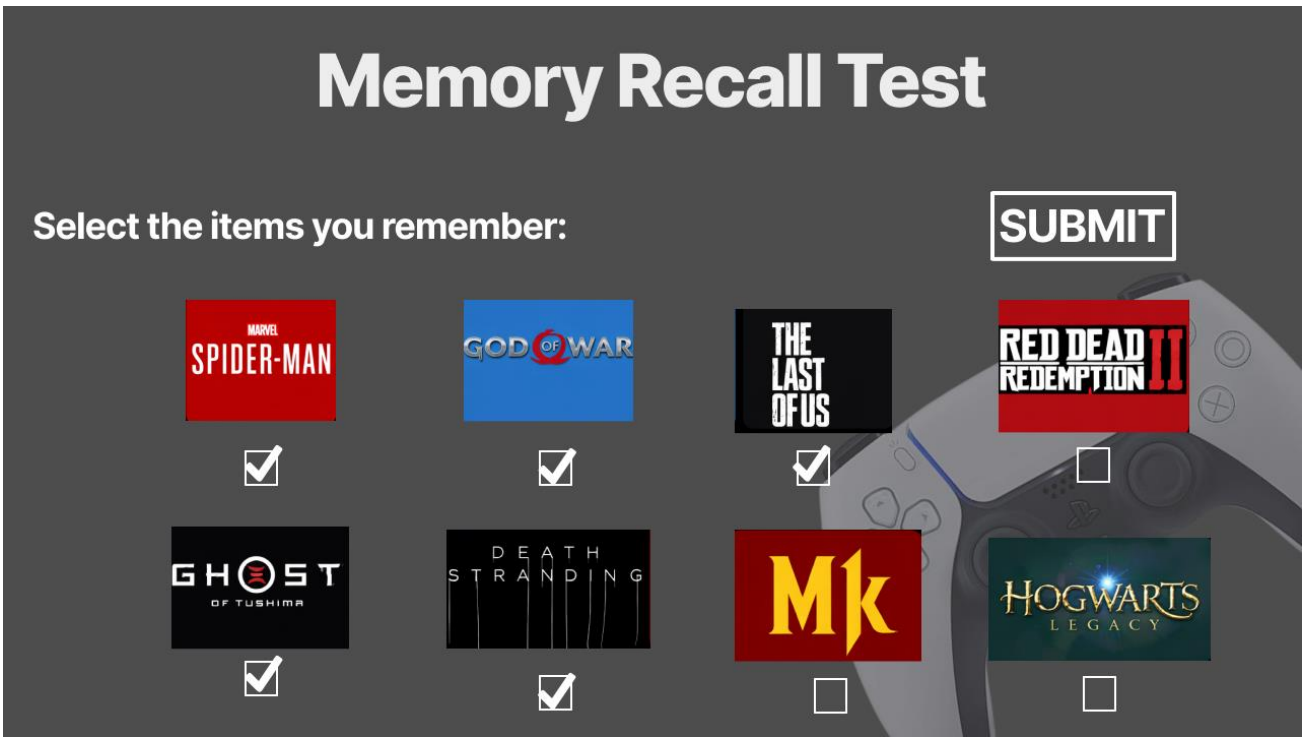
• Enhances short-term memory by allowing users to recall groups of information instead of individual elements.

• Reduces cognitive overload by helping users organize data efficiently.

• Improves pattern recognition, making recall easier and more accurate.

<u>FRAME 3 (RECALL PHASE)</u>:

This screen represents the Selection Phase of a Memory Recall Task, where users recall and choose the items they remember from the previous Chunking Phase. Below is a breakdown of its components:



1. Purpose of the Screen:

• This is the memory retrieval stage, where users select the items they remember seeing in the previous phase.

• The goal is to test the effectiveness of chunking and short-term memory retention.

2. Key Elements and UI Components:

3. How the Selection Phase Works:

• Users analyze the displayed icons and recall which ones they saw in the Chunking Phase.

• They select the remembered icons using the radio buttons below each option.

• Some icons are distractors, testing whether the user's memory is accurate or if they mistakenly recall incorrect icons.

• Clicking "SUBMIT" finalizes their choices, leading to a results or feedback screen.

4. Cognitive and UX Benefits:

• Tests memory accuracy by comparing user selections with previously displayed items.

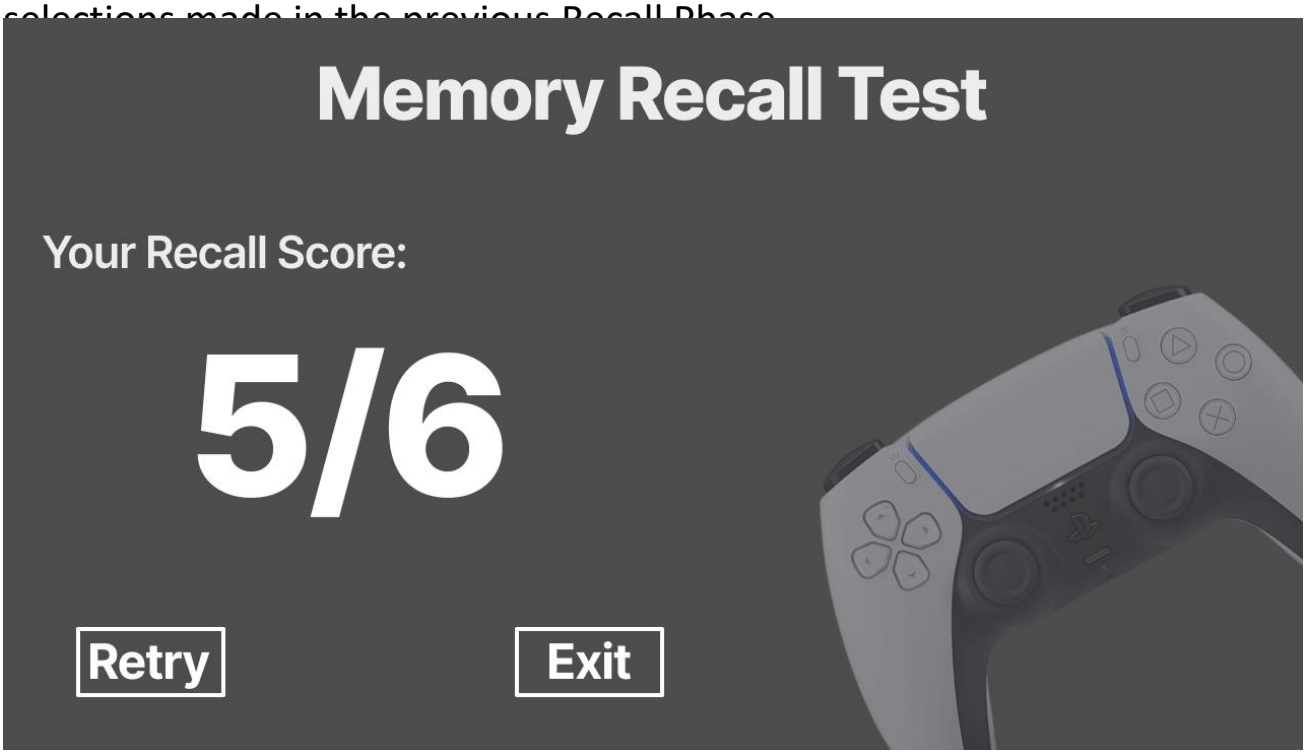
• Incorporates distractors to evaluate how well users distinguish real vs. false memories.

• User-friendly interface with clear selection mechanics (radio buttons).

• Gamified elements (bee character, bright colors) make the task engaging.

<u>FRAME 4 (RESULT PAGE)</u>:

This screen represents the Score & Feedback Phase of the Memory Recall Task, where users receive their performance evaluation based on the selections made in the previous Recall Phase.



1. Purpose of the Screen:

• Provides feedback on recall accuracy by showing the number of correct answers.

• Allows users to decide their next action (continue, restart, or exit).

2. Key Elements & UI Components:

3. How This Phase Works:

• The game evaluates the user's selections from the Recall Phase.

• It calculates the accuracy score (5/6 in this case) and displays it on a wooden board.

• Users review their performance and choose their next action:

  o Continue if they want to proceed.

  o Restart if they want to improve their recall.

  o Exit if they want to stop playing.

4. Cognitive & UX Benefits:

• Instant feedback helps users track their memory performance.

• Multiple options (Continue, Restart, Exit) give users control over their learning experience.

• Visual & gamified elements make the task engaging and less stressful.

LINK:

https://www.figma.com/buzz/p2H7BF2oiHyjSnB0r5YX6o/UID-Exp-2?node-id=0-1&t=35aPU9fhv8gKqd3m-1